

Dr. Tarun Teja Kondraju

Geospatial Scientist | Remote Sensing, UAV & GIS | Ph.D. Spatial Informatics

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PROFILE

Geospatial scientist (Ph.D., IIIT Hyderabad) with **12+ years** of research, teaching and product development in satellite, UAV and field sensing for agriculture, soil, water and the environment. At ICAR-IARI I turn hyperspectral, multispectral, thermal and SAR data into operational tools – including a Google Earth Engine platform for near-real-time crop health monitoring and **DroneMapper**, a desktop drone photogrammetry suite comparable to Pix4D. Co-author of the book **Drone Imaging and Analytics for Agriculture** (ICAR-IARI, 2026) and **24 peer-reviewed publications** (190 citations, h-index 7); experienced teacher and trainer of students, faculty and professionals. Open to roles in academia, research organisations, GIS and remote sensing companies, agriculture and environmental firms, agri-insurance and startups.

12+ yrs

remote sensing & GIS R&D

24 + 1

peer-reviewed papers + book

190

citations · h-index 7

6

geospatial apps built

CORE COMPETENCIES

- Google Earth Engine apps
- Hyperspectral / multispectral / thermal
- Crop & soil parameter mapping
- COLMAP · OpenMVS photogrammetry
- Optical & SAR image processing
- ML & Gaussian process regression
- Water quality remote sensing
- Orthomosaic · DSM/DTM · VI maps
- UAV mission planning & processing
- Radiative-transfer (hybrid) retrieval
- Python · PySide6 desktop apps
- Teaching & capacity building

PROFESSIONAL EXPERIENCE

Research Associate (Post-Doctoral)

Feb 2022 – Present

ICAR – Indian Agricultural Research Institute (IARI), Division of Agricultural Physics, New Delhi

- Conceived and built **PUSAeCMS**, a Google Earth Engine application for near-real-time, field-to-regional crop health monitoring using vegetation, water and composite indices plus physically based biophysical variable retrieval (LAI, canopy chlorophyll).
- Developed further decision-support tools: **PUSA eCMS Desktop** (multi-cloud Sentinel-2 streaming with LAI/CCC/CWC retrieval), **PUSAMIA / EdgeProcessing** (real-time onboard multispectral processing and urea requirement mapping), **VASUDHA** (real-time/offline soil health – N, P, K, OC, EC, pH – from VNIR/SWIR and FTIR-MIR spectra), **PAUDHA** (crop health analytics from spectroradiometer data).
- Plan and execute UAV missions carrying **RGB, multispectral, hyperspectral and thermal** sensors; process imagery end-to-end (Pix4D, WebODM) into analysis-ready crop trait maps.
- Retrieve crop traits (LAI, chlorophyll, nitrogen) using **machine learning, Gaussian process regression and hybrid radiative-transfer models**, scaling from proximal to UAV to Sentinel-2; results published in *Computers and Electronics in Agriculture*, *IIRS* and *Remote Sensing*.
- Developed cloud-based **soil moisture mapping from Sentinel-1 SAR** on Google Earth Engine using regression modelling.
- Co-supervise research scholars in field campaigns, data analysis and manuscript preparation; serve as invited resource person for ICAR drone-technology training programmes.

Senior Research Fellow

Nov 2021 – Feb 2022

Lab for Spatial Informatics, IIIT Hyderabad | DST-funded project

- Satellite-based water quality monitoring (chlorophyll-a) of **Bhadra and Tungabhadra reservoirs** with Sentinel-2 and Google Earth Engine, linking trends to land-use change (published in *Journal of Contaminant Hydrology*, 2024).

GIS Research Consultant & Trainer (Part-time)

Oct 2020 – Oct 2021

AvakAza Geoscience and Research Technologies

- Designed and delivered online training in **ArcGIS and QGIS** for professionals and students from diverse backgrounds, covering spatial analysis workflows for real-world applications.
- Provided GIS research consultancy alongside training delivery.

- Modelled nutrient contamination and water quality in **Nagarjuna Sagar Reservoir** using Landsat and Sentinel-2A/B time series, and coupled it with **SWAT** watershed modelling to quantify fertilizer and precipitation-driven nutrient loading.
- Published first-author work in *Journal of the Indian Society of Remote Sensing*, *ISPRS Annals* and *ISPRS Archives*.
- Research Assistant on the **APN-funded Western Ghats project (OSGeo)** – delivered capacity building in open-source GIS for faculty members from institutions across India.
- Teaching Assistant for undergraduate, postgraduate and research courses: remote sensing data applications, vector and raster processing, spatial analysis, GeoServer and Web-GIS.

PRODUCTS & APPLICATIONS DEVELOPED

- **PUSAeCMS** – Google Earth Engine app for near-real-time crop health management – vegetation/water/composite indices and physically based biophysical variables; quantitative analysis and temporal change visualisation at any location.
- **PUSA eCMS Desktop** – Multi-backend Python/PyQt6 desktop application that streams Sentinel-2 imagery from Microsoft Planetary Computer, Copernicus Data Space and Google Earth Engine (STAC queries, HTTP range requests, resilient retry with exponential backoff) and generates LAI, canopy chlorophyll (CCC) and canopy water content (CWC) maps with Gaussian process regression (scikit-learn, Rasterio, Shapely, PyProj).
- **DroneMapper** – Open-source Windows desktop photogrammetry suite (Python, PySide6) – an alternative to Pix4D and Agisoft Metashape: aligns drone photos (COLMAP), builds dense point clouds and meshes (OpenMVS), and generates georeferenced DSM/DTM, orthomosaics and vegetation-index maps. Supports RGB, multispectral (MicaSense RedEdge/Altum-PT incl. panchromatic and thermal bands) and LWIR thermal cameras with sensor-specific radiometric calibration (downwelling light sensor and calibration panels) and band alignment; built-in 2D/3D viewer, measurement tools, crop-height/phenotyping and processing reports for client delivery.
- **VASUDHA** – Soil health monitoring (N, P, K, OC, EC, pH) from VNIR/SWIR spectroradiometer and FTIR-MIR spectra; real-time or batch offline mode for large spectral datasets.
- **PAUDHA** – Crop health assessment from large spectroradiometer datasets, real-time or offline.
- **PUSAMIA / EdgeProcessing** – Automated edge-computing pipeline for UAV-mounted multispectral cameras (e.g., MicaSense): real-time radiometric calibration to reflectance, SIFT/ORB-based band co-registration and layer stacking, NDVI and crop urea (nitrogen) requirement mapping during flight operations (Python, OpenCV, Rasterio).

INVITED TALKS & RESOURCE PERSON ROLES

- **Guest Lecture: UAV Mission Planning, Data Acquisition (RGB, Multispectral, Hyperspectral, Thermal) and Analytics for Precision Agriculture** – Invited Speaker (18 Dec 2025)
- **Training-cum-Workshop on Drone Technology and Its Application** – ICAR-CIFRI, Barrackpore & ICAR-IARI, New Delhi – Invited Speaker – drone-based imaging spectrometry for precision agriculture (11 Dec 2024)
- **Training Programme on Drone Technology & Its Applications in Agriculture – for scientists, academicians, research scholars and PG/Ph.D. students** – Invited Resource Person – lecture and 8-page training manual chapter (2–3 Dec 2024)

TECHNICAL SKILLS

Programming & Development: Python (PySide6/PyQt6 desktop apps, OpenCV, Rasterio, Shapely, PyProj, scikit-learn), JavaScript/TypeScript, Google Earth Engine app development, STAC APIs (Microsoft Planetary Computer, Copernicus Data Space), COLMAP and OpenMVS, edge computing for UAV payloads, LLM integration

Platforms & Software: Google Earth Engine (app development, big-data processing), ArcGIS, QGIS, ENVI, GeoServer / Web-GIS

UAV & Photogrammetry: Mission planning and data acquisition (RGB, multispectral, hyperspectral, thermal); photogrammetry with Pix4D, WebODM and self-developed DroneMapper (COLMAP, OpenMVS); orthomosaics, DSM/DTM, crop height; LiDAR processing (MicroStation)

Data: Sentinel-1 SAR, Sentinel-2, Landsat, UAV imagery, field spectroradiometry, FTIR spectra

Methods: Machine learning regression, Gaussian process regression, hybrid RTM–ML retrieval, image classification, atmospheric and water-column correction, time-series change analysis, SWAT watershed modelling

Application Domains: Crop health and trait retrieval, soil health and soil moisture, inland water quality, coral reefs, land use / land cover, urban lakes

EDUCATION

Ph.D., Spatial Informatics

2013 – 2021

International Institute of Information Technology (IIIT), Hyderabad

Thesis: Water Quality in Inland Water Bodies and Its Interaction with Land Use (Advisor: Prof. K. S. Rajan)

M.Tech., Energy and Environmental Engineering

2010 – 2012

Vellore Institute of Technology (VIT), Vellore

B.Tech., Biotechnology

2006 – 2010

St. Martin's Engineering College, Hyderabad

SELECTED PUBLICATIONS

1 book (ISBN) + 24 peer-reviewed publications (10 journal, 14 conference/proceedings) | 190 citations, h-index 7 (Google Scholar)
/ complete year-wise list attached separately

1. Sahoo, R. N., **Kondraju, T.**, Ranjan, R., Anand, A., & Rejith, R. G. (2026). *Drone Imaging and Analytics for Agriculture*. ICAR-Indian Agricultural Research Institute, New Delhi. ISBN 978-81-999568-2-7.
2. Sahoo, R. N., Rejith, R. G., **Kondraju, T. T.**, Ranjan, R., Bhandari, A., Gakhar, S., Asim, M., Verrelst, J., Kaur, R., & Singh, T. (2025). Scaling-up plant chlorophyll retrieval from proximal to UAV-borne hyperspectral data using a Gaussian process hybrid model. *International Journal of Remote Sensing*, 1–25.
3. Latwal, A., **Kondraju, T. T.**, Rehana, S., & Rajan, K. S. (2024). Examining chlorophyll-a concentrations in tropical reservoirs under various land-use changes using Sentinel-2 and Google Earth Engine: Bhadra and Tunga-Bhadra, India. *Journal of Contaminant Hydrology*, 265, 104388.
4. Sahoo, R. N., **Kondraju, T. T.**, Rejith, R. G., Verrelst, J., Ranjan, R., Gakhar, S., & Chinnusamy, V. (2024). Monitoring cropland leaf area index using Gaussian process regression and Sentinel-2 surface reflectance data in Google Earth Engine. *International Journal of Remote Sensing*, 45(15), 5008–5027.
5. Sahoo, R. N., Rejith, R. G., Gakhar, S., Verrelst, J., Ranjan, R., **Kondraju, T. T.**, & Chinnusamy, V. (2024). Estimation of wheat biophysical variables through UAV hyperspectral remote sensing using machine learning and radiative transfer models. *Computers and Electronics in Agriculture*, 221, 108942.
6. Sahoo, R. N., Gakhar, S., Rejith, R. G., Verrelst, J., Ranjan, R., **Kondraju, T. T.**, Meena, M. C., Mukherjee, J., Daas, A., & Kumar, S. (2023). Optimizing the retrieval of wheat crop traits from UAV-borne hyperspectral imaging with radiative transfer modelling using Gaussian process regression. *Remote Sensing*, 15(23), 5496.

INTERNATIONAL TRAININGS ATTENDED

- **Training, Validation and Application of Remote Sensing Products in Ecology and Environment for BRICS Countries** – BRICS remote sensing institutions (May 2022)
- **Principles of Ocean Remote Sensing and Its Applications** – ITCOcean & INCOIS, Hyderabad (Jul 2021)

REFERENCES

Available on request.